EuroData-ISRO Internship 2026

FINAL TECHNICAL REPORT

Cross-Modal Satellite Image Retrieval Using Multi-Sensor Remote Sensing Data

EuroSAT RGB • EuroSAT Multispectral • EuroSAT-SAR

Project Item	Details
Competition	ISRO Internship Competition 2026
Problem Statement	Problem 11 – Cross-Modal Satellite Image Retrieval
Modalities	Optical RGB, Sentinel-2 multispectral and Sentinel-1 SAR
Core Tasks	Same-modal retrieval, cross-modal retrieval and land-cover classification
Report Status	Completed using exported experiment metrics and qualitative retrieval figures

Executive Summary

- Developed a **multi-sensor satellite image retrieval framework** for **ISRO Internship Competition 2026 (Problem Statement 11)** using geographically aligned **EuroSAT RGB, Sentinel-2 multispectral, and Sentinel-1 SAR** images.
- Supports **same-modal retrieval** (within RGB, multispectral, and SAR) and **cross-modal retrieval** (RGB↔SAR, RGB↔Multispectral).
- Uses a paired dataset of **27,000 samples per modality** across **10 land-use/land-cover classes**, with a **70:15:15 stratified train/validation/test split** (18,900 / 4,050 / 4,050).
- Preserves identical sample IDs and split assignments across modalities to **avoid pair mismatches and cross-modal data leakage**.
- Achieved **98.20% test accuracy and weighted F1-score** for the multispectral classifier; RGB models reached **95.33–95.53%**, while SAR achieved **83.26%** due to speckle noise and higher scene ambiguity.
- Retrieval performance was strongest for **multispectral-to-multispectral** and **multispectral-to-optical** searches, while **SAR-based retrieval** remained the most challenging.

1. Dataset Information

1.1 Dataset Composition

The project combines three aligned representations of the EuroSAT land-cover benchmark. EuroSAT-RGB contains three visible Sentinel-2 channels. EuroSAT multispectral retains all 13 Sentinel-2 bands. EuroSAT-SAR provides matched Sentinel-1 dual-polarization VV and VH patches. A shared file identifier links corresponding RGB, multispectral and SAR samples into a tri-modal index.

Dataset / Modality	Sensor	Channels / Bands	Format	Patch Size	Samples	Classes
EuroSAT-RGB (Optical)	Sentinel-2	3 (RGB)	JPEG	64×64	27,000	10
EuroSAT Multispectral	Sentinel-2	13 bands	GeoTIFF	64×64	27,000	10
EuroSAT-SAR	Sentinel-1	2 (VV, VH)	GeoTIFF	64×64	27,000	10
Paired tri-modal index	S1 + S2	Shared file IDs	CSV metadata	Aligned	27,000 triplets	10

1.2 Class Distribution and Shared Stratified Split

Class ID	Class Name	Total	Train 70%	Validation 15%	Test 15%
0	AnnualCrop	3,000	2,100	450	450
1	Forest	3,000	2,100	450	450
2	HerbaceousVegetation	3,000	2,100	450	450
3	Highway	2,500	1,750	375	375
4	Industrial	2,500	1,750	375	375
5	Pasture	2,000	1,400	300	300
6	PermanentCrop	2,500	1,750	375	375
7	Residential	3,000	2,100	450	450
8	River	2,500	1,750	375	375
9	SeaLake	3,000	2,100	450	450
—	Total	27,000	18,900	4,050	4,050

3. Model and Evaluation Framework

Each modality is processed by its own encoder and projection head. The projection vectors are L2-normalized and compared using cosine similarity. Classification heads encourage the embeddings to preserve land-cover semantics, while triplet and alignment objectives bring matching or semantically related samples together in the common representation space.

Component	Method
Modality encoders	Separate optical, multispectral and SAR feature extractors.
Embedding head	Dense projection followed by L2 normalization.
Classification head	Ten-class softmax branch for semantic supervision.
Metric learning	Triplet loss for relative similarity structure.
Cross-modal alignment	Alignment loss for paired modality embeddings.
Ranking	Descending cosine similarity between query and gallery embeddings.
Relevance criterion	A gallery item is relevant when its land-cover label matches the query label.
Reported K values	K = 1, 5, 10, 20 and 50.

3.1 Retrieval Metric Interpretation

Metric	Interpretation
Precision@K	Proportion of the Top-K retrieved images that are relevant.
Recall@K	Proportion of all relevant gallery images recovered within the Top-K.
F1@K	Harmonic mean of Precision@K and Recall@K.
Classification precision	Among predictions assigned to a class, the proportion that is correct.
Classification recall	Among true samples of a class, the proportion correctly detected.
Classification F1-score	Balanced summary of class-level precision and recall.

3.2 Training Behaviour

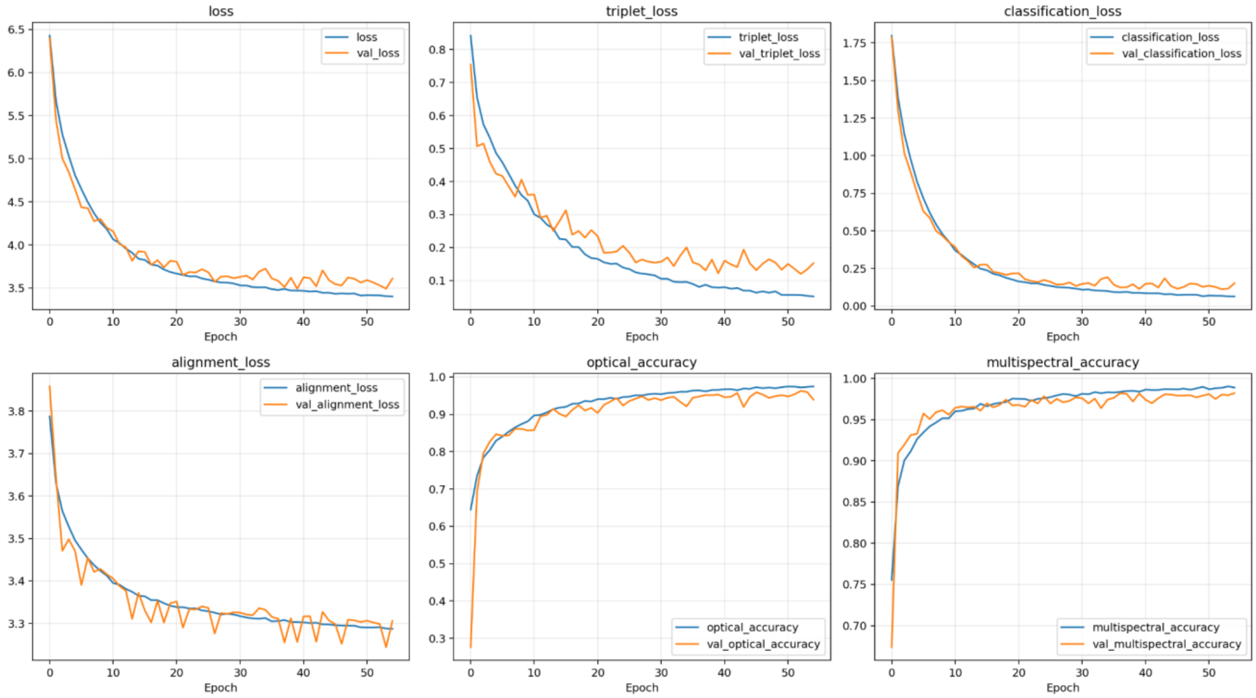

Figure 1. Optical–multispectral training history: total, triplet, classification and alignment losses with branch accuracies.

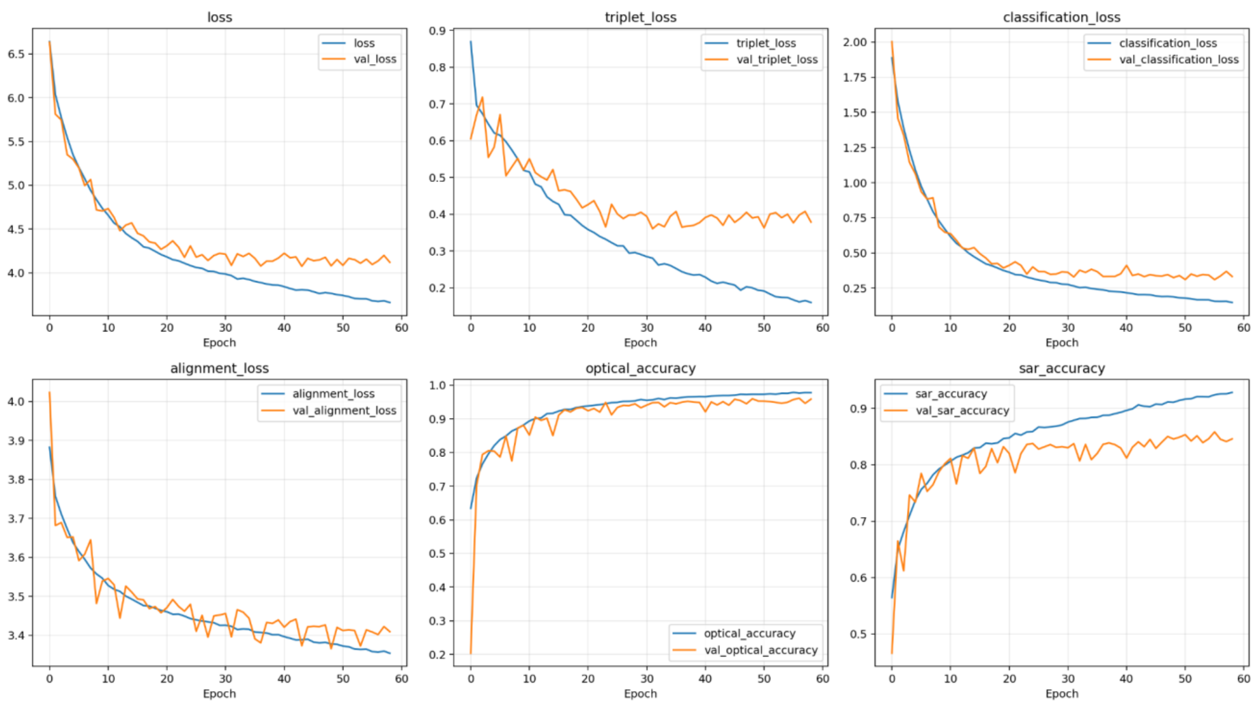

Figure 2. Optical–SAR training history: total, triplet, classification and alignment losses with optical and SAR accuracies.

4. Classification Results

4.1 Optical Classification – Optical/SAR Model

Class	Precision	Recall	F1-score	Support
AnnualCrop	0.9546	0.9356	0.9450	450
Forest	0.9719	0.9978	0.9846	450
HerbaceousVegetation	0.9278	0.9133	0.9205	450
Highway	0.9915	0.9387	0.9644	375
Industrial	0.9806	0.9413	0.9605	375
Pasture	0.9148	0.9667	0.9400	300
PermanentCrop	0.9098	0.8880	0.8988	375
Residential	0.9594	0.9978	0.9782	450
River	0.9410	0.9787	0.9595	375
SeaLake	0.9911	0.9867	0.9889	450
Accuracy	0.9553	0.9553	0.9553	—
Macro Average	0.9543	0.9544	0.9540	4,050
Weighted Average	0.9556	0.9553	0.9552	4,050

4.2 SAR Classification

Class	Precision	Recall	F1-score	Support
AnnualCrop	0.8106	0.8844	0.8459	450
Forest	0.7858	0.9378	0.8551	450
HerbaceousVegetation	0.6184	0.8067	0.7001	450
Highway	0.9103	0.7040	0.7940	375
Industrial	0.9448	0.9120	0.9281	375
Pasture	0.6698	0.7100	0.6893	300
PermanentCrop	0.7424	0.3920	0.5131	375
Residential	0.9391	0.9600	0.9495	450
River	0.9749	0.9333	0.9537	375
SeaLake	0.9844	0.9800	0.9822	450
Accuracy	0.8326	0.8326	0.8326	—
Macro Average	0.8381	0.8220	0.8211	4,050
Weighted Average	0.8402	0.8326	0.8277	4,050

SAR performs strongly on Residential, River, SeaLake and Industrial scenes. PermanentCrop is the most difficult category, with recall of 0.3920 and F1-score of 0.5131, suggesting substantial overlap with other vegetation classes in radar backscatter patterns.

4.3 Optical Classification – Optical/Multispectral Model

Class	Precision	Recall	F1-score	Support
AnnualCrop	0.8935	0.9511	0.9214	450
Forest	0.9844	0.9822	0.9833	450
HerbaceousVegetation	0.9158	0.9422	0.9288	450
Highway	0.9678	0.9627	0.9652	375
Industrial	0.9706	0.9680	0.9693	375
Pasture	0.9493	0.9367	0.9430	300
PermanentCrop	0.8886	0.9147	0.9014	375
Residential	0.9977	0.9578	0.9773	450
River	0.9856	0.9120	0.9474	375
SeaLake	0.9889	0.9911	0.9900	450
Accuracy	0.9533	0.9533	0.9533	—
Macro Average	0.9542	0.9518	0.9527	4,050
Weighted Average	0.9545	0.9533	0.9536	4,050

4.4 Multispectral Classification

Class	Precision	Recall	F1-score	Support
AnnualCrop	0.9627	0.9756	0.9691	450
Forest	0.9934	0.9978	0.9956	450
HerbaceousVegetation	0.9954	0.9644	0.9797	450
Highway	0.9810	0.9627	0.9717	375
Industrial	0.9867	0.9893	0.9880	375
Pasture	0.9340	0.9900	0.9612	300
PermanentCrop	0.9731	0.9653	0.9692	375
Residential	0.9956	0.9978	0.9967	450
River	0.9839	0.9760	0.9799	375
SeaLake	1.0000	0.9978	0.9989	450
Accuracy	0.9820	0.9820	0.9820	—
Macro Average	0.9806	0.9817	0.9810	4,050
Weighted Average	0.9823	0.9820	0.9820	4,050

Multispectral classification is the strongest branch. Every class has an F1-score above 0.96, and the weighted F1-score is 0.9820. The additional spectral bands improve separation among vegetation and built-up categories.

5. Same-Modal Retrieval Results

5.1 SAR → SAR Retrieval

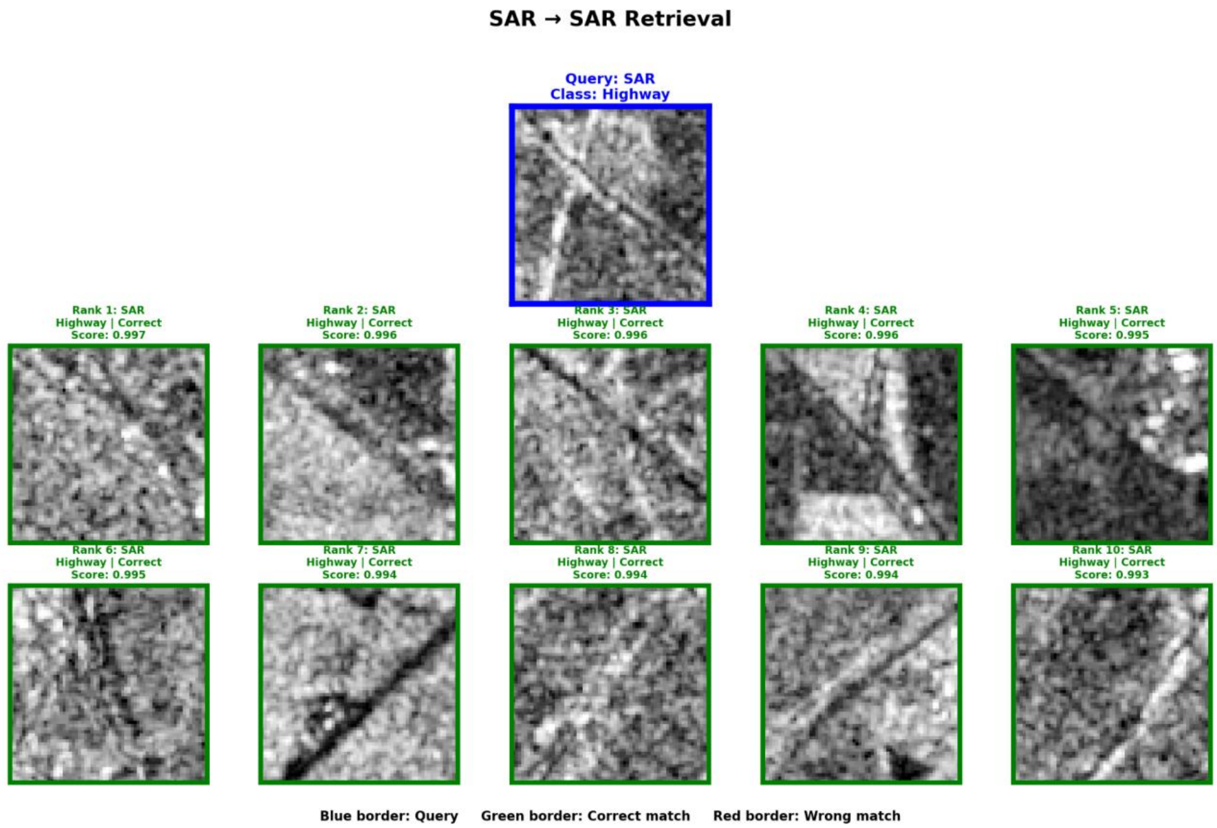

Figure 3. SAR query and Top-10 SAR retrieval results for the Highway class.

K	Precision@K	Recall@K	F1@K
1	0.8111	0.0020	0.0040
5	0.8043	0.0099	0.0195
10	0.8043	0.0198	0.0386
20	0.8038	0.0395	0.0752
50	0.8024	0.0985	0.1752

Precision remains close to 0.80 across K. The qualitative example retrieves ten correct Highway scenes, showing that individual high-confidence queries can perform better than the dataset-wide average.

5.2 Optical → Optical Retrieval

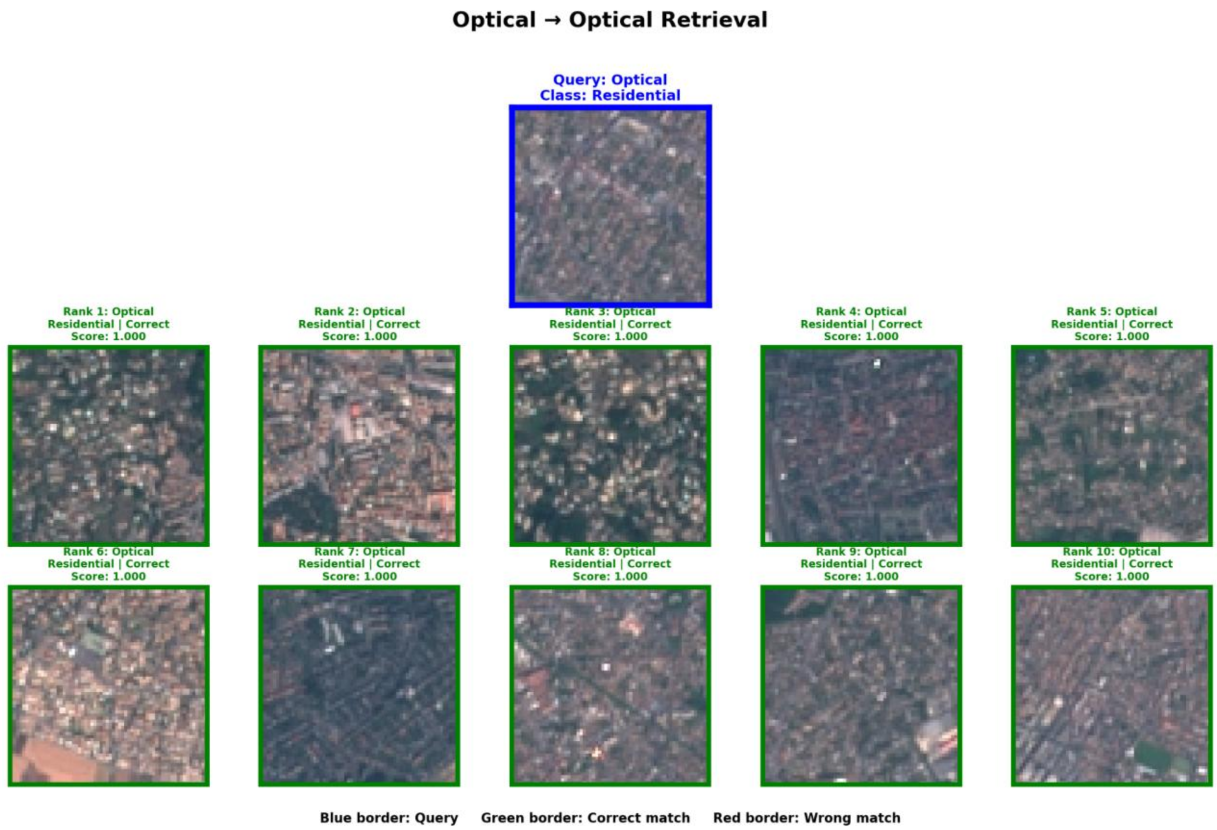

Figure 4. Optical query and Top-10 optical retrieval results for the Residential class.

K	Precision@K	Recall@K	F1@K
1	0.9548	0.0024	0.0047
5	0.9528	0.0118	0.0233
10	0.9514	0.0235	0.0459
20	0.9507	0.0470	0.0895
50	0.9509	0.1175	0.2088

Optical same-modal precision remains approximately 0.95 from K=1 to K=50. Recall rises with K because more relevant gallery samples are included, while precision remains stable.

5.3 Multispectral → Multispectral Retrieval

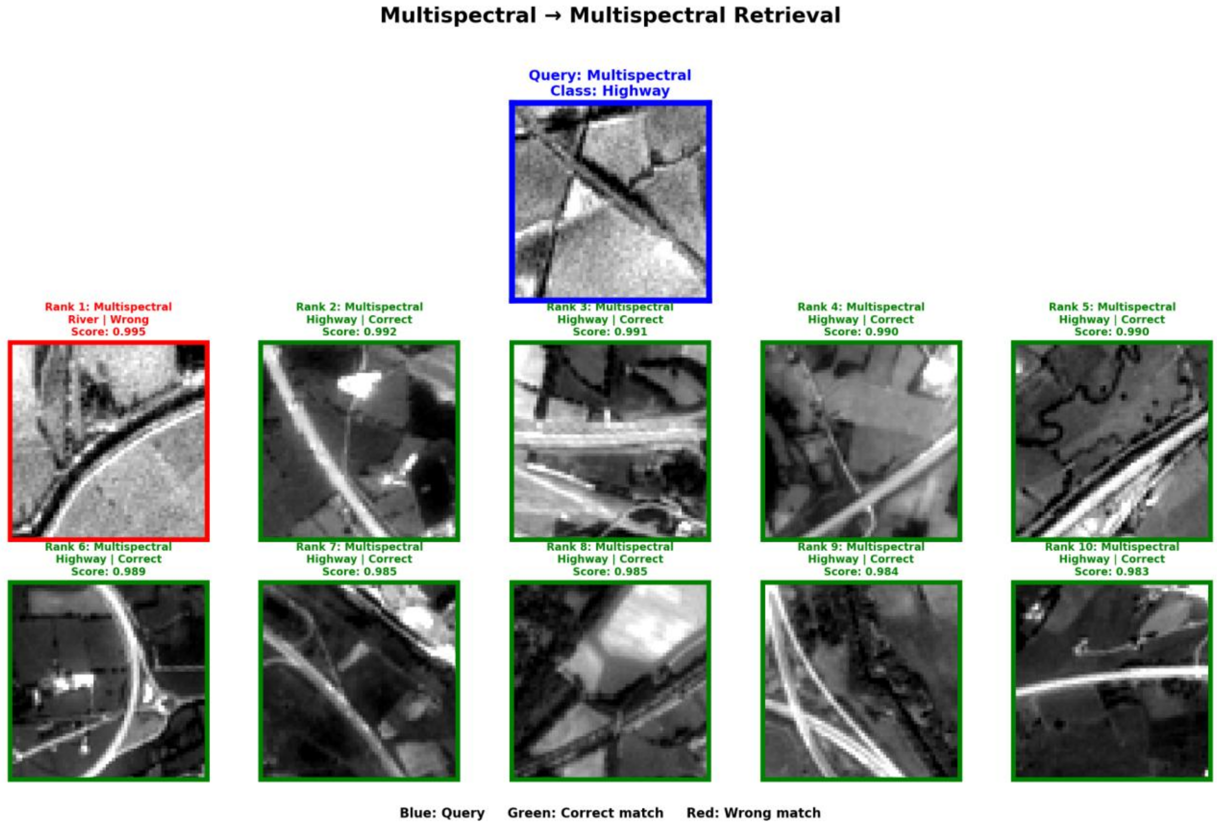

Figure 5. Multispectral query and Top-10 multispectral retrieval results for the Highway class.

K	Precision@K	Recall@K	F1@K
1	0.9768	0.0024	0.0048
5	0.9779	0.0121	0.0239
10	0.9781	0.0242	0.0472
20	0.9790	0.0484	0.0922
50	0.9799	0.1212	0.2154

This is the strongest same-modal direction, with Precision@1 of 0.9768 and Precision@50 of 0.9799. The displayed query has one River false positive at rank 1, followed by nine correct Highway results.

6. Cross-Modal Retrieval Results

6.1 Optical → SAR

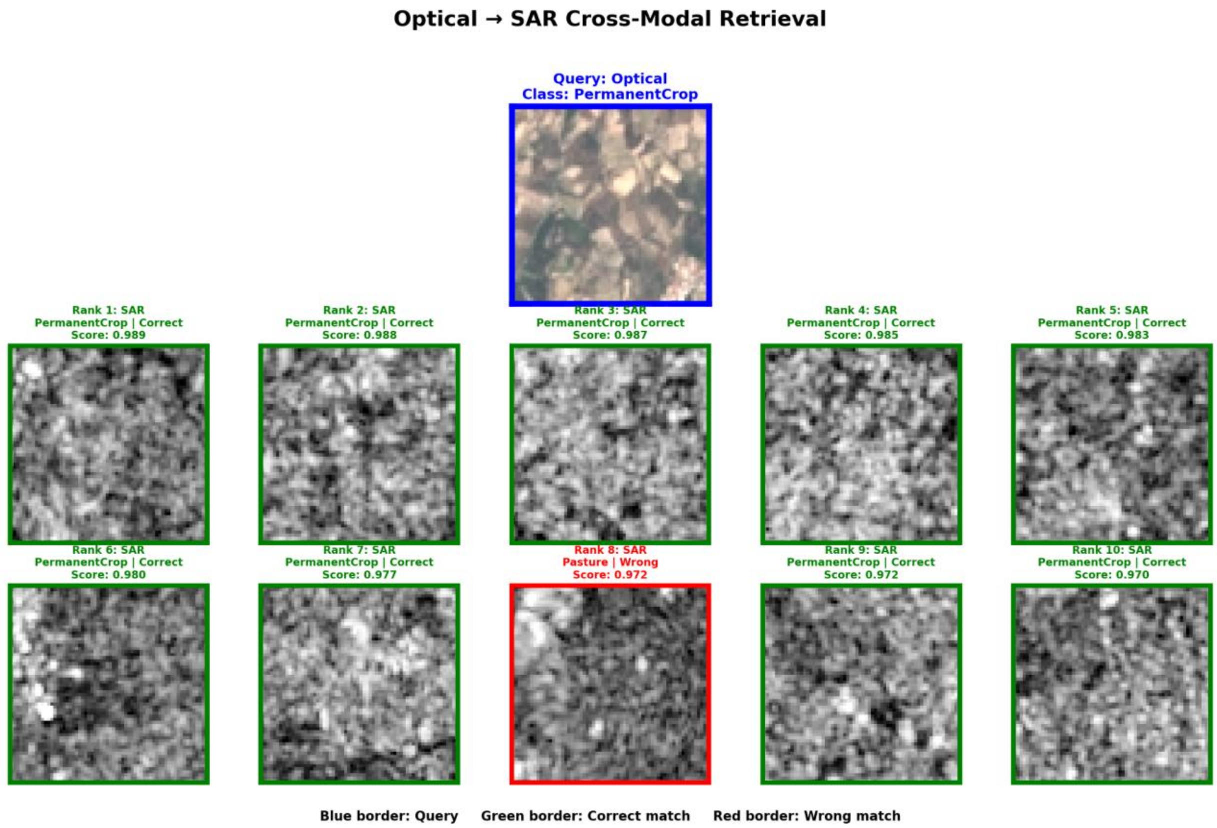

Figure 6. Optical PermanentCrop query with Top-10 SAR results.

K	Precision@K	Recall@K	F1@K
1	0.9262	0.0023	0.0046
5	0.9302	0.0115	0.0226
10	0.9268	0.0228	0.0445
20	0.9261	0.0456	0.0869
50	0.9195	0.1133	0.2014

The model achieves Precision@1 of 0.9262 and maintains precision above 0.919 through K=50. In the qualitative example, nine of ten retrieved SAR images are relevant; the only error is a Pasture sample at rank 8.

6.2 SAR → Optical

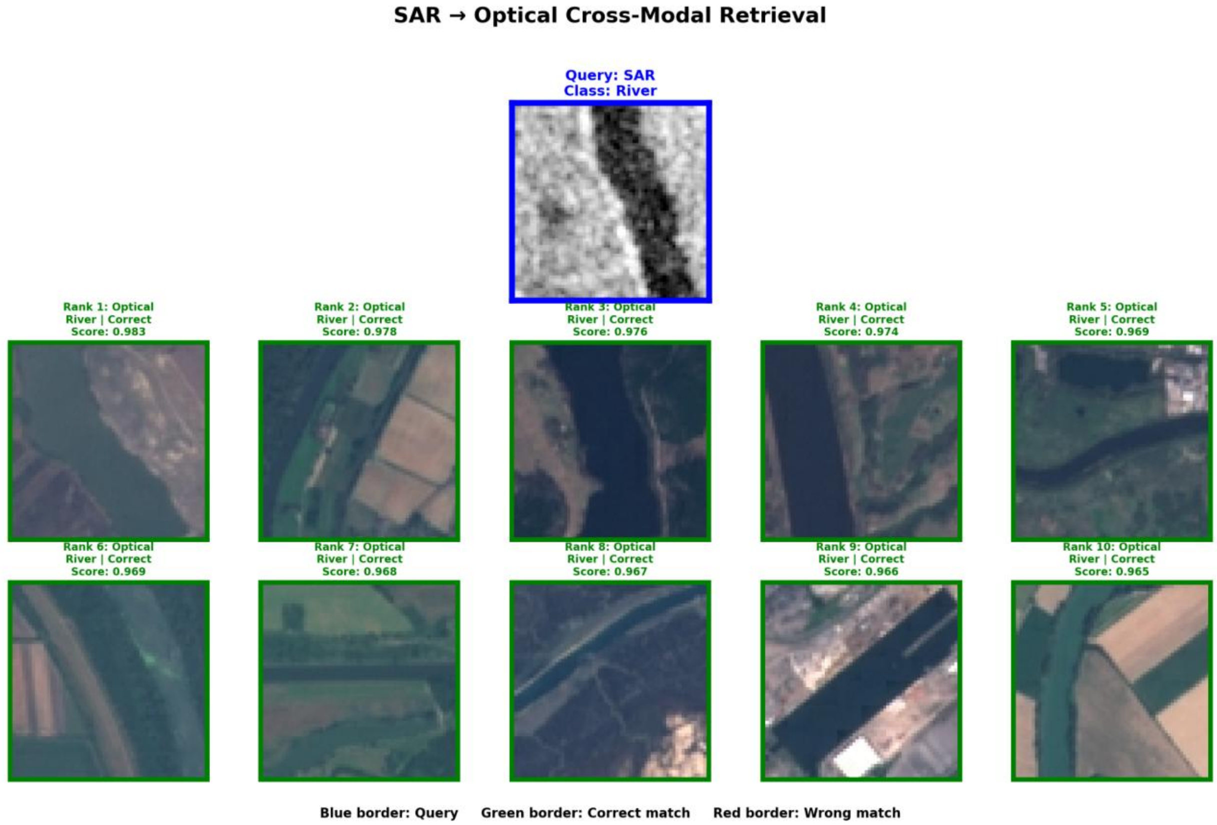

Figure 7. SAR River query with Top-10 optical results.

K	Precision@K	Recall@K	F1@K
1	0.7963	0.0019	0.0039
5	0.8053	0.0098	0.0193
10	0.8055	0.0195	0.0382
20	0.8071	0.0392	0.0747
50	0.8094	0.0984	0.1752

SAR-to-optical retrieval is more difficult than the reverse direction, with Precision@1 of 0.7963. Nevertheless, the selected River query retrieves ten correct optical River scenes, illustrating good performance for visually distinctive classes.

6.3 Optical → Multispectral

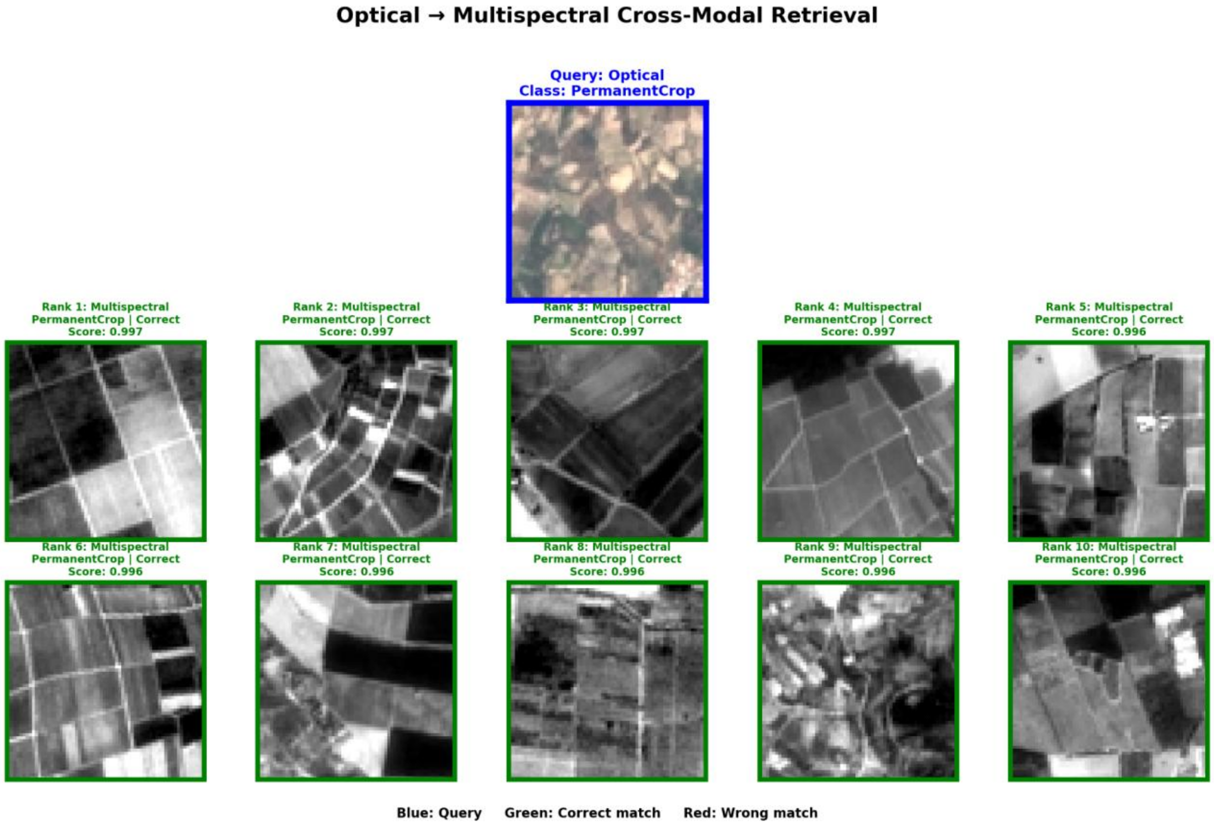

Figure 8. Optical PermanentCrop query with Top-10 multispectral results.

K	Precision@K	Recall@K	F1@K
1	0.9432	0.0023	0.0046
5	0.9388	0.0115	0.0228
10	0.9411	0.0232	0.0452
20	0.9434	0.0464	0.0885
50	0.9469	0.1166	0.2073

Precision ranges from 0.9388 to 0.9469 across the tested K values. The qualitative example returns ten correct PermanentCrop images, showing effective semantic alignment from visible RGB to full multispectral data.

6.4 Multispectral → Optical

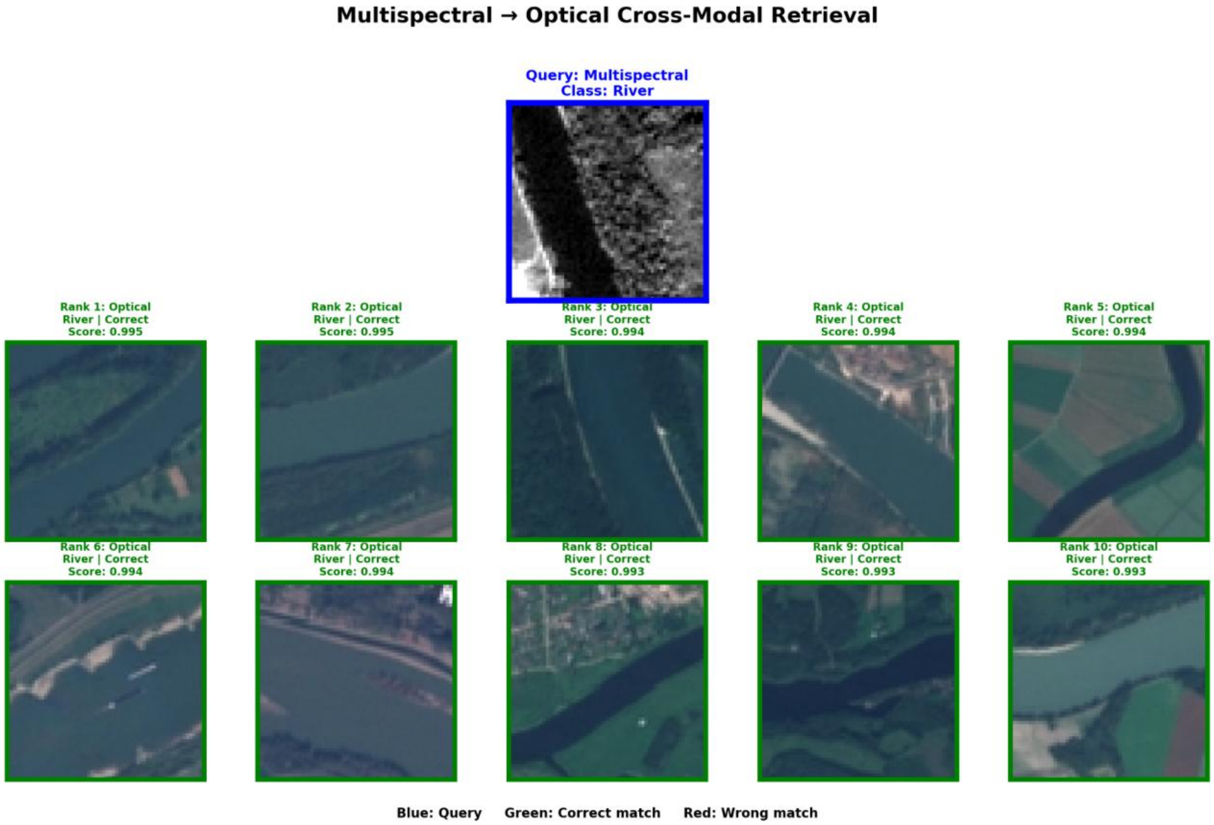

Figure 9. Multispectral River query with Top-10 optical results.

K	Precision@K	Recall@K	F1@K
1	0.9716	0.0024	0.0048
5	0.9728	0.0120	0.0237
10	0.9740	0.0240	0.0469
20	0.9755	0.0482	0.0917
50	0.9776	0.1207	0.2144

This is the strongest cross-modal direction, with Precision@1 of 0.9716 and Precision@50 of 0.9776. The selected River query retrieves ten correct optical images.

7. Discussion

7.1 Classification Summary

Modality / Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Optical (Optical/SAR)	0.9553	0.9543	0.9544	0.9540	0.9552
SAR	0.8326	0.8381	0.8220	0.8211	0.8277
Optical (Optical/MS)	0.9533	0.9542	0.9518	0.9527	0.9536
Multispectral	0.9820	0.9806	0.9817	0.9810	0.9820

7.2 Precision@K Across Retrieval Directions

Direction	P@1	P@5	P@10	P@20	P@50
Optical → to → Optical	0.9548	0.9528	0.9514	0.9507	0.9509
SAR → to → SAR	0.8111	0.8043	0.8043	0.8038	0.8024
Multispectral → to → Multispectral	0.9768	0.9779	0.9781	0.9790	0.9799
Optical → to → SAR	0.9262	0.9302	0.9268	0.9261	0.9195
SAR → to → Optical	0.7963	0.8053	0.8055	0.8071	0.8094
Optical → to → Multispectral	0.9432	0.9388	0.9411	0.9434	0.9469
Multispectral → to → Optical	0.9716	0.9728	0.9740	0.9755	0.9776

7.3 Recall@K Across Retrieval Directions

Direction	R@1	R@5	R@10	R@20	R@50
Optical → to → Optical	0.0024	0.0118	0.0235	0.0470	0.1175
SAR → to → SAR	0.0020	0.0099	0.0198	0.0395	0.0985
Multispectral → to → Multispectral	0.0024	0.0121	0.0242	0.0484	0.1212
Optical → to → SAR	0.0023	0.0115	0.0228	0.0456	0.1133
SAR → to → Optical	0.0019	0.0098	0.0195	0.0392	0.0984
Optical → to → Multispectral	0.0023	0.0115	0.0232	0.0464	0.1166
Multispectral → to → Optical	0.0024	0.0120	0.0240	0.0482	0.1207

7.4 F1@K Across Retrieval Directions

Direction	F1@1	F1@5	F1@10	F1@20	F1@50
Optical → to → Optical	0.0047	0.0233	0.0459	0.0895	0.2088
SAR → to → SAR	0.0040	0.0195	0.0386	0.0752	0.1752
Multispectral → to → Multispectral	0.0048	0.0239	0.0472	0.0922	0.2154
Optical → to → SAR	0.0046	0.0226	0.0445	0.0869	0.2014
SAR → to → Optical	0.0039	0.0193	0.0382	0.0747	0.1752
Optical → to → Multispectral	0.0046	0.0228	0.0452	0.0885	0.2073
Multispectral → to → Optical	0.0048	0.0237	0.0469	0.0917	0.2144